HVAC BIZ PRIVATE LIMITED
Class-1 Licensed HVAC Turnkey Contractor & Enterprise Software

PROPRIETARY VERTICAL ERP PLATFORM

Transforming High-Stakes HVAC & MEP Contracting into a High-Margin Science

Connecting commercial bidding, parametric BOQ engineering, real-time site procurement, milestone billing, and statutory accounting into a single unified digital thread.

₹3.19 Cr+	24.3%	100%	Zero
Active Project Portfolio	Target Gross Margin	BOQ 3-Way Reconciliation	Spreadsheet Data Leakage

The $120B MEP Contracting Crisis

Why Generic ERPs (SAP, Odoo, QuickBooks) Fail Specialty Engineering Contractors

⚠ PAINFUL REALITY FOR 85,000+ CONTRACTORS

A Disconnected Stack Bleeds 6% to 8% of Contractor Margin on Every Job

Specialized contractors handle complex bills of quantities, multi-tiered subcontractor labor, OEM equipment orders, and rigorous site commissioning. Generic software fails to bridge the gap between engineering plans and field reality.

AVERAGE PROJECT EROSION ANATOMY:
Initial Quoted Margin: **19.2%** → Final Handover Margin: **8.4%** (Over 56% of profits lost to untracked variances, unbilled scope creep, and procurement delays).

Spreadsheet Estimation Silos

Estimators build complex quotes in Excel. When the tender is awarded, line items are never mapped to warehouse inventory or procurement codes, causing immediate ordering blunders.

Uncontrolled Site Procurement

Field engineers purchase expensive copper pipes, refrigerants, and valves locally at premium retail rates because central warehouse availability is opaque.

Delayed Milestone Cashflow & Retention Drag

Contractors wait months to bill clients because milestone completion evidence is scattered across WhatsApp chats, paper challans, and fragmented emails.

Compliance & Payroll Vulnerability

Managing field technician site attendance, statutory PF/ESIC deductions, and GST e-invoicing manually creates severe audit and regulatory exposure.

The Solution: AlphaERP Ecosystem

A Single Source of Truth from First Site Survey to Final Handover Certificate

Engineered Ground-Up for HVAC, VRF, Chiller Plants & Electrical Infrastructure

Unlike generic accounting software or lightweight CRMs, AlphaERP links every commercial line item directly to engineering specifications, live warehouse inventory, and milestone receivables.

1

Precision BOQ Engineering

Dynamic line-item costing with automatic margin calculations. Instantly converts winning proposals into active project budgets with 1-click conversion.

Client PDF Generator Included

2

Controlled Supply Chain

Automated site shortages trigger Indents (PR) and 3-way matched Purchase Orders (PO). Zero unauthorized off-book purchases, preventing margin erosion.

Real-time Warehouse Staging

3

Automated Finance & HR

Milestone-based progress billing with retention aging. Integrated double-entry General Ledger, GST/TDS tax records, and 12-subsystem AlphaHRMS payroll.

Full Statutory Compliance

Experience the Entire Working Ecosystem
11 integrated core modules built and operating live today.

SYSTEM DESIGN • Product Showcase

Platform Architecture & Digital Thread

How Data Flows Seamlessly Across Every Operational Department

1. COMMERCIAL HUB

CRM & Quotations

Leads, Site Surveys, Parametric BOQ, Client PDF Generator.

2. ENGINEERING CORE

Projects & BOQ Engine

Material Allocation, Budget Milestones, Variance & Variance Alerts.

3. SUPPLY CHAIN

PR/PO & Multi-Warehouse

Site Indents, 3-Way Match, Vendor PO Dispatch, SKU Stock.

4. FINANCIALS & HR

Ledger, Billing & HRMS

Progress Invoicing, Retention Aging, Journal Vouchers, Payroll.

DIGITAL THREAD EXECUTION PIPELINE

Zero Manual Double-Entry

Tender Lead	Quotation BOQ	Active Project	PR / PO Dispatch	Milestone Bill
Heat-Load Calc	Cost vs Selling	Auto Material Sync	3-Way Verification	Ledger + GST TDS

Security: Role-Based Access Control (RBAC: Admin, Project Manager, Site Engineer, Finance, HR)

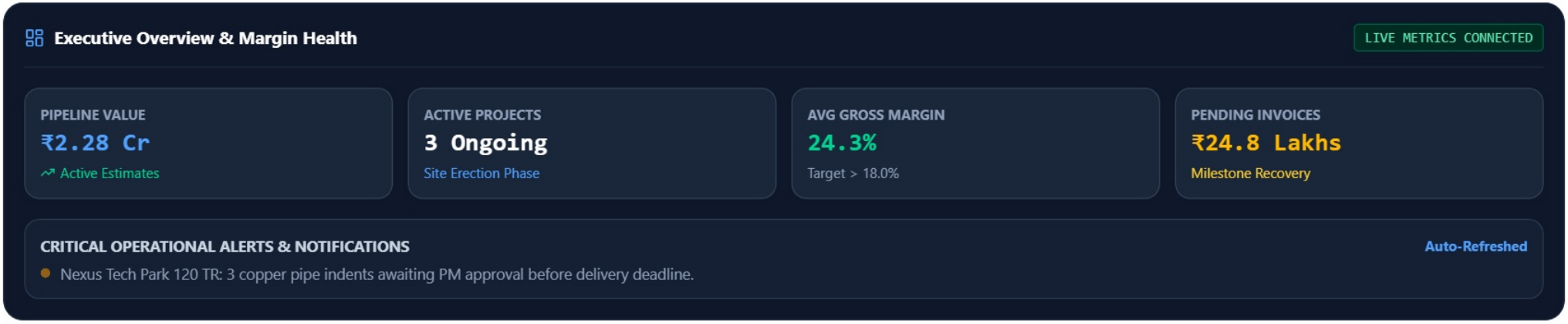
Cloud Sync: Google Sheets 2-Way API

SCREEN WALKTHROUGH • Product Showcase

Main Screen 1: Executive Command Center

Live Operational KPIs, Revenue Digest, Margin Sparklines & Critical Alerts

Module: **src/components/DashboardView.tsx**



- Real-time Gross Margin Sparklines
- Bi-directional Google Sheets API Sync
- Role-Based Privacy Masking for Staff

SCREEN WALKTHROUGH • Product Showcase

Main Screen 2: CRM & HVAC Leads Pipeline

Managing Complex Commercial Inquiries, Heat-Load Surveys & Tenders

Module: **src/components/CRMLeadsView.tsx**

HVAC Pipeline Stages & Technical Site Surveys — 5 Enterprise Inquiries

Nexus Enterprises	Lodha FinTech Hub	Hiranandani Corporate
Site Survey Done	Quotation Issued	Won & Awarded
Scope: 120 TR VRF Heat Pump	Scope: 45 TR Chilled Water Fan Coil	Scope: 80 TR High-Sensible IT Hub
Est: ₹1.08 Cr	Est: ₹42.5 Lakhs	Est: ₹78.9 Lakhs
Rep: Vikramaditya Solanki	Rep: Sneha Kulkarni	Project Converted: PRJ-2026-003

Integrated WhatsApp & Email proposal transmission logged with audit trails. **1-Click "Award to Project"**

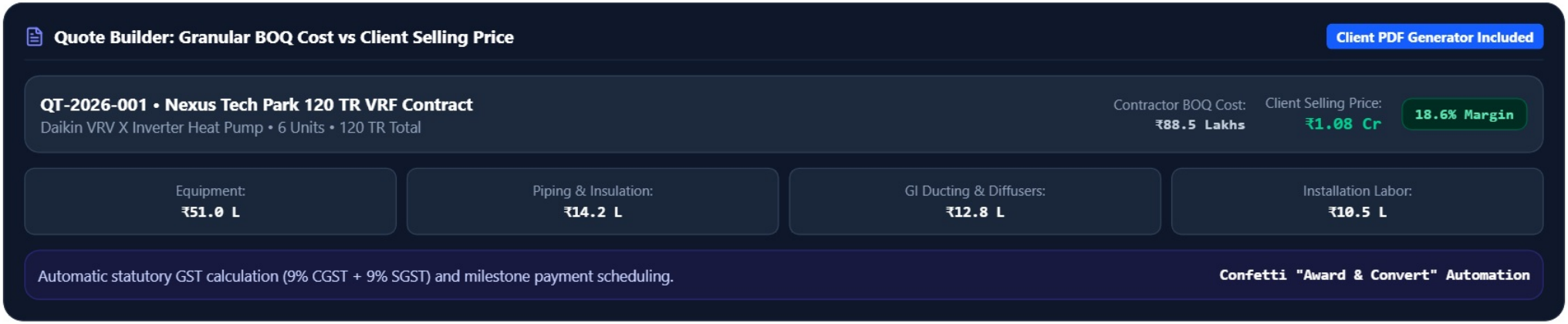
- Heat-Load Calculation Records
- Sales Rep Commission Attribution
- WhatsApp Direct Quote Dispatch

SCREEN WALKTHROUGH • Product Showcase

Main Screen 3: Quotes & BOQ Estimations

Granular Cost vs Selling Price, Margin Guardrails & Client PDF Generator

Module: **src/components/QuotesView.tsx**

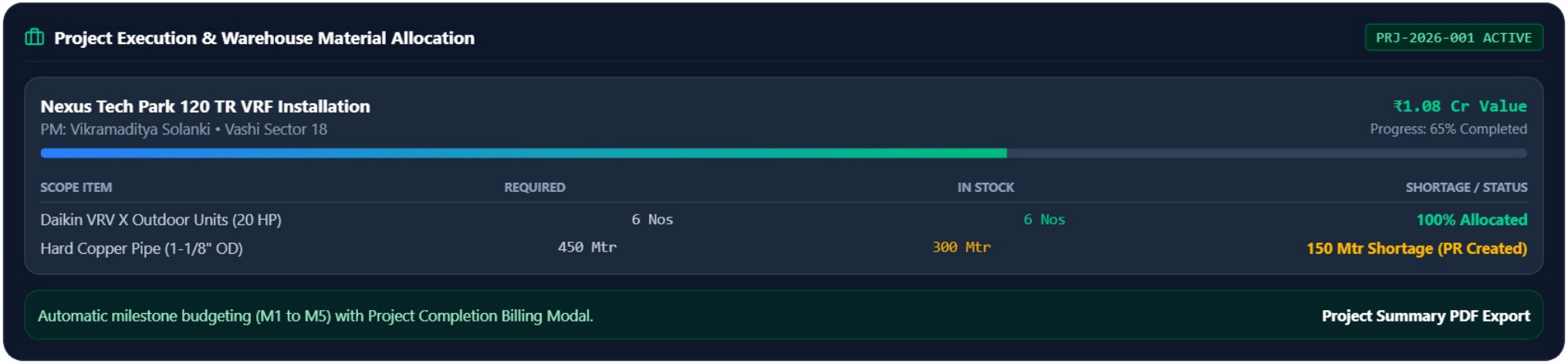


- Real-time Gross Margin Guardrails
- Letterhead Client PDF Generation
- Seamless Conversion to Project Execution

Main Screen 4: Projects & BOQ Inventory Engine

Execution Tracking, Milestone Budgets & Real-Time Material Shortages

Module: **src/components/ProjectsBOQView.tsx**



- Real-time Shortage Detection
- Auto Purchase Requisition Trigger
- Completion Bill & Retention Ledger

SCREEN WALKTHROUGH • Product Showcase

Main Screen 5: Project Cost & PO Variance

Target Budget vs Committed POs vs Actuals: Stopping Margin Leakage Early

Module: **src/components/ProjectCostAnalysisView.tsx**

Target Cost vs Committed PO vs Actual Spend Breakdown

PROFIT MARGIN GUARDRAIL

ESTIMATED COST (BUDGET)	COMMITTED POS (PURCHASED)	PROJECTED NET MARGIN
₹88.50 Lakhs	₹86.20 Lakhs	20.1% (+1.5%)
Baseline Quote	-₹2.30 L Savings	Above Quoted 18.6%

Category Level Variance Tracking:

Equipment variance: **0.0%** • Piping & Insulation: **-2.4% (Favorable)** • Ducting & Grilles: **+1.8% (Raw GI Steel Spike)**.

- Real-time Margin Erosion Alerts
- Committed PO vs Uncommitted Delta
- Early Warning Before Invoice Clearance

SCREEN WALKTHROUGH • Product Showcase

Main Screen 6: Procurement PR & PO Pipeline

3-Way Matching, Multi-Tier Approvals & Instant Vendor PO Generation

AlphaERP
Series A Presentation

Module: **src/components/ProcurementPRPOView.tsx**

Purchase Requisition (PR) to Approved PO Flow

3 Active Purchase Orders

PR-2026-004 — Pending PM Approval

Copper Tubing & Armacell Class-O Insulation

Requested by: Site Eng. Rohan • Project: Nexus 120 TR

Est. Amount: ₹3.42 Lakhs

PO-2026-001 — Issued to Vendor

Daikin Airconditioning India Pvt Ltd

Item: 6x VRV X Heat Recovery ODUs + Y-Joints

PO Value: ₹41.40 Lakhs (GST Inc.)

Automatic 3-Way Match between PO, Goods Receipt Note (GRN), and Vendor Invoice.

PO PDF Document Export

- Multi-Tier Approval Chain
- Vendor Rate Benchmarking
- Automated PO Delivery Tracking

SCREEN WALKTHROUGH • Product Showcase

Main Screens 7 & 8: Warehouse & Field Operations

Multi-Location Stock Control, Serial Tracking & Mobile Field Ticketing

Modules: **InventoryWarehouseView & WorkOrdersView**

Warehouse Stock & Reorder Triggers

Multi-Location

R-410A Refrigerant Cylinders (45kg)	18 In Stock
VRF Inverter Compressor 10HP	2 Units (Reorder Level)
Galvanized Steel Sheets (22 Gauge)	4.8 Tons In Stock

Barcode/QR scanning for site dispatch & staging.

Field Work Orders & Commissioning

Mobile Ready

WO-901: Nitrogen Pressure Testing	Passed (450 PSI)
WO-902: Air Balancing (TAB) CFM Signoff	In Progress
WO-903: Company Tools Return Verification	Flaring Tool Returned

Includes company asset return logging & remark tracking.

- Minimum Stock Safety Thresholds
- Technician Digital Signatures
- Asset Return Remarks & Checklist

SCREEN WALKTHROUGH • Product Showcase

Main Screens 9 & 10: Invoicing, Receivables & General Ledger

Milestone Progress Billing, Retention Recovery, TDS & Double-Entry Accounting

Modules: **FinancePaymentsView & FinanceView**

Contractor Financial Integrity & Statutory Tax Schedule

Tally / SAP Interface Ready

PROGRESS MILESTONE BILLING

INV-2026-003

₹32.40 Lakhs (30% Equipment Dispatch)

Status: Client Paid via RTGS

RETENTION MONEY LEDGER

₹10.80 Lakhs (10%)

Matures: 12 Mos Post Handover

Automated Aging Notification

TAX COMPLIANCE SCHEDULE

18% GST (Works Contract)

CGST 9% + SGST 9% Reconciled

TDS 2% Section 194C Deducted

Automated Double-Entry Journal Vouchers generated on every invoice approval & PO settlement. **Balance Sheet & P&L Ready**

- Milestone Receivables Aging
- Automated Bank RTGS Coordinates
- Statutory TDS & GST Vouchers

SCREEN WALKTHROUGH • Product Showcase

Main Screen 11: AlphaHR Suite & Employee Portal

ATS Recruitment, Biometric Attendance, Statutory Payroll & 3-Tier Exit Approvals

Module: **src/components/HRView.tsx**

AlphaHR: 12 Integrated Subsystems & Self-Service Portal

12 Enterprise Subsystems

1. Talent Acquisition
Kanban ATS & Candidate Pipeline

2. Attendance & Roster
Biometric Geo-fenced Punch

3. Statutory Payroll
PF, ESIC, PT, TDS Formula Engine

4. Multi-Tier Exit Flow
Manager → Admin → HR → Finance F&F

Full & Final (F&F) Settlement Automation
Tracks company asset handover remarks, notice pay, gratuity, and leaves.

Zero Audit Lag

- Automated Payslip PDF Generation
- ESS / MSS Mobile Self-Service
- Statutory Compliance Guarantee

Market Opportunity & Expansion

$250 Billion Global HVAC Boom Powered by Urbanization & Data Centers

TOTAL ADDRESSABLE MARKET

$250 B+

Global HVAC equipment installation & turnkey MEP engineering services by 2030.

SERVICEABLE ADDRESSABLE MARKET

$45 B

Commercial & industrial HVAC contractors in India, Middle East, and Southeast Asia.

SERVICEABLE OBTAINABLE MARKET

$1.8 B

Vertical SaaS & ERP software spend by Tier-1 & Tier-2 engineering contractors.

MACRO GROWTH TAILWINDS ACCELERATING ADOPTION:

1. Data Center Boom
Precision liquid and chilled water cooling contracts growing at 28% CAGR.

2. Green Building Mandates
LEED & IGBC compliance requires strict verification of equipment efficiency & air balancing.

3. Generational Handover
Second-generation contractor leadership demands modern cloud ERP over outdated desktop software.

MONETIZATION • Market & Business

Business Model & Unit Economics

SaaS Subscription Tiers + High-Margin Enterprise Implementation

Growth Contractor

₹25,000 / mo

For contractors executing up to 5 concurrent projects.

- CRM & Lead Pipeline
- BOQ Quote Estimator
- Client PDF Generator
- 10 Team User Seats

MOST POPULAR

Turnkey Enterprise

₹65,000 / mo

For mid-size MEP contractors handling up to 25 active sites.

- Everything in Growth
- PR/PO 3-Way Match & Warehouse
- Milestone Progress Invoicing
- Full AlphaHRMS & Payroll
- 50 User Seats

EPC Mega-Projects

₹1,50,000+ / mo

For large industrial infrastructure & data center contractors.

- Unlimited Active Projects
- Custom General Ledger COA
- Dedicated Cloud Run Tenant
- Bi-directional ERP Bridge
- 24/7 Priority SLA Support

Target LTV/CAC: **5.8x**

Gross Margin on Software: **84%**

Net Revenue Retention: **124%**

Competitive Moat: Why We Win

Generic Horizontal ERPs vs Specialized Vertical Mastery

CAPABILITY	ALPHAERP	SAP / ORACLE	ODOO / ZOHO	EXCEL / SPREADSHEETS
Turnkey HVAC BOQ Cost vs Selling Builder	✓ Yes	✕ No	✕ No	✓ Manual
Automated Client Quotation PDF Generator	✓ Yes	✕ No	✕ No	✕ No
Live Material Shortage to PR Auto-Trigger	✓ Yes	✕ No	✕ No	✕ No
Project Cost Variance & Margin Guardrail	✓ Yes	✓	✕ No	✕ No
Retention Money Aging & Progress Invoicing	✓ Yes	✕ No	✕ No	✕ No
Integrated 12-Subsystem AlphaHRMS Payroll	✓ Yes	✕ No	✕ No	✕ No
Time to Value / Deployment Speed	2 Weeks	12-18 Months	3-6 Months	Immediate
Implementation Cost	Low	$150k+	$25k+	Hidden (Errors)

Core Moat: Real-world operational battle-testing by HVAC Biz ensures every workflow, unit rate, and approval logic aligns with practical site reality.

Vision, Product Roadmap & The Ask

Building the Global Operating System for MEP Infrastructure Engineering

Q3 - Q4 2026

AI Drawing BOQ Takeoff

Computer vision & LLM parser to ingest 2D DWG and PDF floor plans, automatically extracting duct lengths, tonnage, and diffuser counts.

Q1 - Q2 2027

IoT BMS Telematics

Direct integration with Daikin, Voltas, and Carrier BACnet controllers for continuous sensor monitoring and predictive warranty maintenance.

Q3 - Q4 2027

Supplier Marketplace

Direct wholesale material ordering from certified copper, steel, and insulation manufacturers with automated trade credit financing.

CONTACT & INVESTOR RELATIONS

HVAC BIZ PRIVATE LIMITED

Plot No. 42, Sector 18, Vashi Commercial Hub, Navi Mumbai, MH - 400705

Email: contact@hvacbiz.com • Phone: +91 22 2789 4400